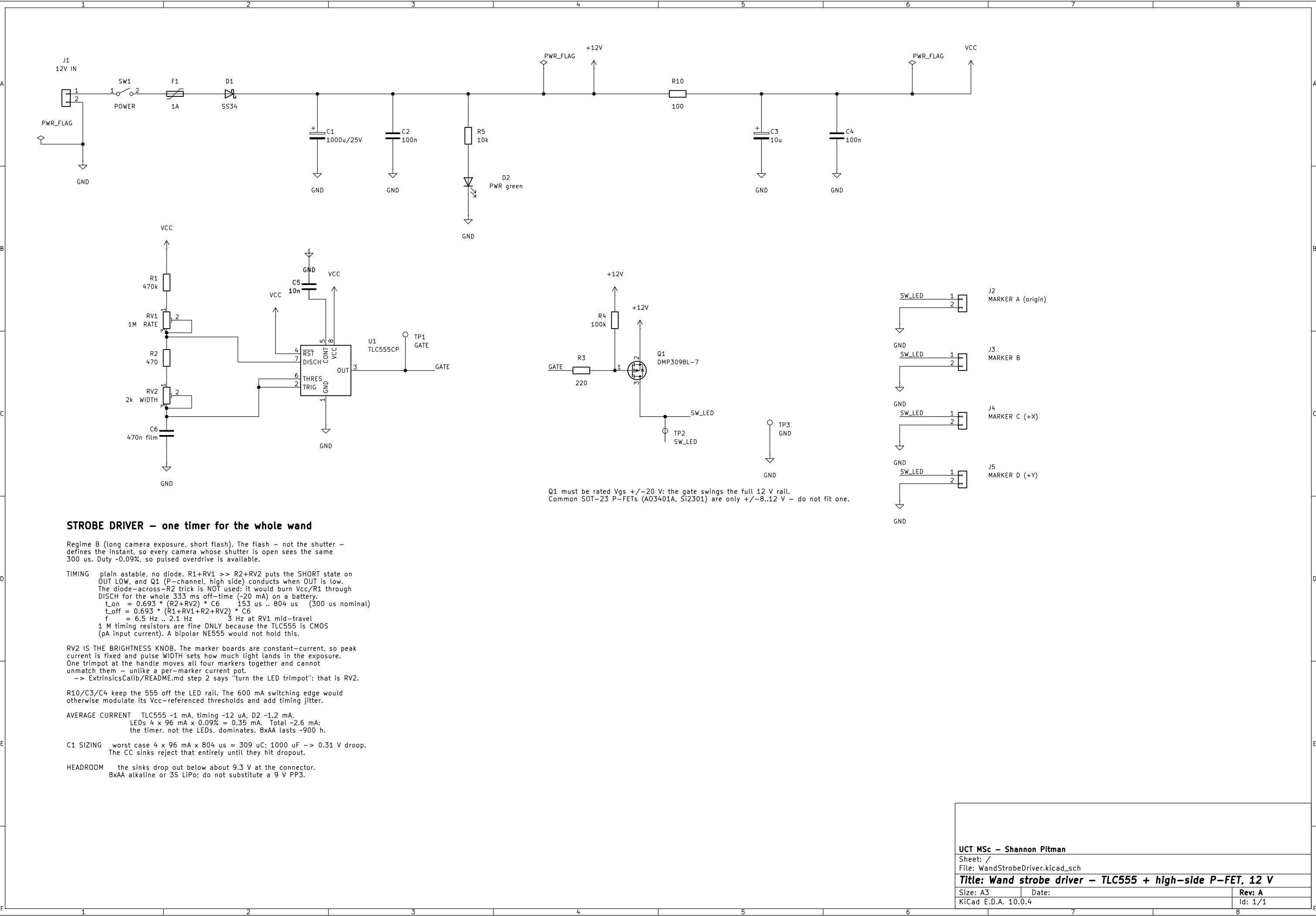




STROBE DRIVER – one timer for the whole wand

Regime B (long camera exposure, short flash). The flash – not the shutter – defines the instant, so every camera whose shutter is open sees the same 300 us. Duty ~0.09%, so pulsed overdrive is available.

TIMING plain astable, no diode. $R1+RV1 \gg R2+RV2$ puts the SHORT state on OUT LOW, and Q1 (P-channel, high side) conducts when OUT is low. The diode-across-R2 trick is NOT used; it would burn Vcc/R1 through DISCH for the whole 333 ms off-time (~20 mA) on a battery.
 $t_{on} = 0.693 * (R2+RV2) * C6$ 153 us .. 804 us (300 us nominal)
 $t_{off} = 0.693 * (R1+RV1+R2+RV2) * C6$
 $f = 6.5 \text{ Hz} \dots 2.1 \text{ Hz} \dots 3 \text{ Hz}$ at RV1 mid-travel
1 M timing resistors are fine ONLY because the TLC555 is CMOS (pA input current). A bipolar NE555 would not hold this.

RV2 IS THE BRIGHTNESS KNOB. The marker boards are constant-current, so peak current is fixed and pulse WIDTH sets how much light lands in the exposure. One trimpot at the handle moves all four markers together and cannot unmatch them – unlike a per-marker current pot.
-> ExtrinsicCalib/README.md step 2 says "turn the LED trimpot": that is RV2.

R10/C3/C4 keep the 555 off the LED rail. The 600 mA switching edge would otherwise modulate its Vcc-referenced thresholds and add timing jitter.

AVERAGE CURRENT TLC555 ~1 mA, timing ~12 uA, D2 ~1.2 mA, LEDs $4 \times 96 \text{ mA} \times 0.09\% = 0.35 \text{ mA}$. Total ~2.6 mA: the timer, not the LEDs, dominates. 8xAA lasts ~900 h.

C1 SIZING worst case $4 \times 96 \text{ mA} \times 804 \text{ us} = 309 \text{ uC}$; 1000 uF -> 0.31 V droop. The CC sinks reject that entirely until they hit dropout.

HEADROOM the sinks drop out below about 9.3 V at the connector. 8xAA alkaline or 3S LiPo; do not substitute a 9 V PP3.

UCT MSc – Shannon Pitman

Sheet: /
File: WandStrobeDriver.kicad_sch

Title: Wand strobe driver – TLC555 + high-side P-FET, 12 V

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